Chapter 1

Frequency Map-Making

Contents

Introduction

In Chapter 4, we presented the frequency dependency of QUBIC's beam, and the possibility of using this property, called *spectral imaging*, to split the physical bandwidth into frequency sub-bands to increase the spectral resolution of the instrument. We now introduce a map-making algorithm based on spectral imaging, to reconstruct multiple frequency maps of the sky. We call this algorithm *Frequency Map-Making* (FMM). The FMM method is similar to imagers' map-making algorithms, which are naturally designed to reconstruct numerous frequency maps, in contrast with CMM (presented in Chapter 6) which is more specific to BI. We start by introducing the general fundamentals of map-making for CMB experiments, followed by the QUBIC-specific formalism covering both the forward simulation model and the FMM reconstruction algorithm. The resulting maps are then analysed to understand the behaviour of FMM : numerical cost, noise structure and anti-correlations introduced by spectral imaging. This chapter remains methodological, deeper analysis and forecasts are presented in the subsequent chapters of this thesis. It is important to note that the entire map-making effort relies on data simulation using the instrumental model developed in the Chapter 8; the methods described in this chapter and the next have not yet been applied to real sky data, only on real calibration source data in laboratory [1, 2].

The content of this chapter is the result of multiple people within QUBIC Collaboration. In 2014, the master internship of Federico Incardona on the understanding of the polychromatic form of the synthesized beam generated the idea of the polychromatic reconstruction. The first trial of implementation was done the next year by the intern student Gohar Dashyan, and finalised during Mikhail Stolpovsky thesis [3]. Further tests were performed during the thesis of Louise Mousset [4]. Mathias Regnier and I continued these efforts by building a robust FMM pipeline and deeper analysis of this method [5], leading to the paper [6].

Personal contribution: During my first year, I contributed with Mathias Regnier to the development and the implementation of the Frequency Map-Making algorithm. This work is presented in Mathias' thesis [5], and led to a publication [6]. During my second year, I completely refactored and cleaned the FMM code, then I built a pipeline allowing simulation, reconstruction and analysis for FMM, easy to use from a single parameters file. During my third year, I extended the analysis of the numerical parameters and their effect on the reconstruction, leading to a forecast-ready pipeline for QUBIC instrument. My contributions to FMM are further developed in Chapters 7, 8 and 9.

1.1 Map-Making: Concept & Formalism

Map-making algorithms lie at the heart of the data analysis pipeline of all CMB telescopes. These algorithms use Time-Ordered Data (TOD), datasets in which each sample corresponds to the sky signal measured by the instrument at a given time, to reconstruct maps of the sky. This reconstruction relies on the pointing direction of the telescope at each time sample, together with a model of the instrument. In this section, we develop the standard map-making formalism used by most CMB experiments and its adaptation to the specific features of QUBIC.

1.1.1 Map-Making Solution

The purpose of map-making is to reconstruct sky maps from telescope data. In CMB experiments, the data are commonly referred to as Time-Ordered Data (TOD), since the temporal dependence of the measurements is a key element in the reconstruction process through the scanning strategy of the instrument. This differs significantly from typical telescopes which rely on long exposure observations to produce accurate and low-noise images of the sky. In contrast, CMB instruments continuously scan the sky following specific observing strategies. Different scanning strategies present different advantages and drawbacks, but their primary purpose is to mitigate systematic effects, such as electronic or cryogenic fluctuations, by observing the same sky location under different instrumental conditions. Scanning strategies also help reduce the impact of atmospheric $1/f$ noise, which dominates at low frequencies. In addition, modern CMB experiments aim to measure the polarisation of the CMB. This requires observing the same region of the sky with different detector orientation angles in order to distinguish the Stokes parameters Q and U (see 1.3.3).

The most general way to describe the data acquisition at one time is :

$$d(\vec{u}(t)) = \int_{\nu_{min}}^{\nu_{max}} \mathcal{H}_\nu(\vec{u}(t)) s_\nu d\nu + n(\vec{u}(t)), \tag{1.1}$$

where $\vec{u}(t)$ is the pointing direction on the sky at time t, d is the TOD at time t, ν_{min} and ν_{max} the physical frequency bandwidth of the instrument, $\mathcal{H}_\nu$ is the acquisition operator of the instrument, which models the data acquisition of the instrument (including convolution, PSF, optics, electronics, etc), s_ν is the sky map, and n is the detector and photon noise contribution at time t. $\mathcal{H}_\nu$ is a very complex object, which will be discussed in detail in Chapter 8.

For simplicity, the scanning strategy is included in the acquisition operator. The entire TOD acquisition can then be written as :

$$\begin{aligned} \vec{d} &= \int_{\nu_{min}}^{\nu_{max}} \mathcal{H}_\nu \vec{s_\nu} d\nu + \vec{n} \\ &= H\vec{s} + \vec{n}, \end{aligned} \tag{1.2}$$

defining H as the acquisition operator integrating into the frequency band. The goal is now to estimate the unknown sky $\vec{s}$ of size (N_{pix}, N_{stokes}), N_{stokes} being equal to 3 for I, Q, U parameters (see subsection 1.3.3), from the data $\vec{d}$ and the instrument model encoded in the acquisition operator H. Assuming Gaussian noise $\vec{n}$, all its statistical information is stored in the noise covariance matrix $N = \langle \vec{n}\vec{n}^{\,T} \rangle$. Its probability distribution, noting $N_{\text{pointings}}$ the number of time samples, is then written :

$$P(\vec{n}) = \frac{1}{|(2\pi)^{N_{\text{pointings}}} N|^{1/2}} \exp\left(-\frac{1}{2} \vec{n}^{\,T} N^{-1} \vec{n} \right). \tag{1.3}$$

The best solution for map-making equation 1.2 is obtained by maximising the likelihood function giving the probability of getting data $\vec{d}$ from a sky $\vec{s}$, noted $\mathcal{L} \equiv P(\vec{d}|\vec{s})$, as detailed in [7]. Using Bayes theorem [8] :

$$\mathcal{L} \equiv P(\vec{d}|\vec{s}) = \frac{P(\vec{s}|\vec{d}) P(\vec{d})}{P(\vec{s})} \propto P(\vec{s}|\vec{d}), \tag{1.4}$$

where the last proportionality assumes a flat (uninformative) prior $P(\vec{s}) = \text{const}$.

Since the noise is Gaussian, $\mathcal{L} \propto P(\vec{n})$. From equation 1.2, $\vec{n} = \vec{d} - H\vec{s}$. The likelihood function can then be written as :

$$\mathcal{L} \propto P(\vec{n}) = \frac{1}{|(2\pi)^{N_{\text{pointings}}} N|^{1/2}} \exp\left(-\frac{1}{2} (\vec{d} - H\vec{s})^{\,T} N^{-1} (\vec{d} - H\vec{s}) \right). \tag{1.5}$$

Maximising the likelihood $\mathcal{L}$ corresponds to computing the most likely sky $\vec{s}$ to correspond to the data $\vec{d}$ according to the noise $\vec{n}$. Equivalently, we can minimise the χ^2 function associated with this likelihood :

$$\chi^2 \equiv -2 \ln \mathcal{L} \propto (\vec{d} - H\vec{s})^{\,T} N^{-1} (\vec{d} - H\vec{s}). \tag{1.6}$$

To minimise this function, we need to compute the sky for which the derivative of χ^2 is zero. The minimisation proceeds as follows:

$$\begin{aligned}
&\frac{\partial \chi^2}{\partial \vec{s}} = \frac{\partial \chi^2}{\partial \vec{s}^{\,T}} = 0 \\
\Longleftrightarrow\ &\frac{\partial \chi^2}{\partial \vec{s}^{\,T}} = \frac{\partial}{\partial \vec{s}^{\,T}}(\vec{d}^{\,T}N^{-1}\vec{d} - \vec{d}^{\,T}N^{-1}H\vec{s} - \vec{s}^{\,T}H^{\,T}N^{-1}\vec{d} + \vec{s}^{\,T}H^{\,T}N^{-1}H\vec{s}) = 0 \\
&\Rightarrow -H^{\,T}N^{-1}\vec{d} + H^{\,T}N^{-1}H\vec{\hat{s}} = 0 \\
&\Rightarrow \vec{\hat{s}} = (H^{\,T}N^{-1}H)^{-1}H^{\,T}N^{-1}\vec{d},
\end{aligned} \tag{1.7}$$

where we note $\vec{\hat{s}}$ the maximal-likelihood solution of the map-making equation.

Unfortunately, despite $\vec{\hat{s}}$ having an analytical solution, it requires to compute the inverse of the matrix $M \equiv (H^{\,T}N^{-1}H)$. M has a size $(N_{pix} \times N_{pix})$, with $N_{pix} \sim 10^7$ for modern experiments. Inversion algorithms, like Cholesky factorisation or LU decomposition, have a computation cost in $\mathcal{O}(N_{pix}^3)$ and a memory usage in $\mathcal{O}(N_{pix}^2)$ [9]. It then requires $\sim 10^{21}$ FLOP to invert such a matrix. With the CPU on my laptop[1], being able to perform around 100 GFLOP/s, I would need many years to perform an inversion ($\sim$ 320 years with Cholesky decomposition). Even with exascale supercomputer, with 10^{18} FLOP$/s$, $\sim$ 20 minutes are still needed. But, the main issue comes from memory, as in double precision[2], the computation needs $8 \times (N_{pix})^2 = 800$ TB, just to store the matrix.

From these elements, matrix inversion is not practically feasible on realistic CMB datasets.

1.1.2 Iterative Solver: PCG

To overcome the difficulty of matrix inversion, modern map-making algorithms rely on iterative solvers, such as the Preconditioned Conjugate Gradient (PCG) method, which avoid explicit matrix inversion and only require repeated evaluations of the operator. Indeed, equation 1.7 can be rewritten :

$$\begin{aligned}
\vec{\hat{s}} &= (H^{\,T}N^{-1}H)^{-1}H^{\,T}N^{-1}\vec{d} \\
\Rightarrow (H^{\,T}N^{-1}H)\vec{\hat{s}} &= H^{\,T}N^{-1}\vec{d} \\
\Longleftrightarrow A\vec{x} &= \vec{b},
\end{aligned} \tag{1.8}$$

with $A \equiv (H^{\,T}N^{-1}H)$ and $b \equiv H^{\,T}N^{-1}\vec{d}$. This linear equation is solvable using a Conjugate Gradient (CG), which will minimise the function $f(\vec{x}) = \frac{1}{2}\vec{x}^{\,T}A\vec{x} - \vec{b}^{\,T}\vec{x} + c$ [10], with c a constant. The gradient of f is :

$$\nabla f(\vec{x}) = A\vec{x} - \vec{b}. \tag{1.9}$$

CG algorithm will iterate on $\vec{x}$ by searching for the solution along a set of mutually conjugate search directions to minimise $f(\vec{x})$, which exactly corresponds to finding the

[1] Intel i7-13850HX

[2] 8 bytes per element

solution of equation 1.8. A complete description of PCG algorithm can be found in [11]. Still, the size of $\vec{x}$ remains a problem, at the order of $N_{pix} \sim 10^7$. The preconditioner transforms the linear system into an equivalent one that converges faster, and can be seen as a matrix that approximates A while being easier to invert, such that $M^{-1}A$ has a more clustered eigenvalue spectrum than A itself. A common preconditioner used in CMB experiments is $diag(A)$.

1.2 Numerical Implementation & Methodology

We now extend this formalism to the specific case of QUBIC. Two complementary operations must be distinguished. The forward model (simulation) computes synthetic TOD, noted $\tilde{\vec{d}}$, from a known input sky $\tilde{\vec{s}}$ using equation 1.2; this is used to validate the algorithm and study its properties under controlled conditions. The reconstruction does the reverse: given $\tilde{\vec{d}}$, the PCG-based map-making algorithm of Section 1.1 recovers an estimated sky $\hat{\vec{s}}$. Throughout this chapter, we use tildes ($\tilde{\vec{s}}$, $\tilde{\vec{d}}$) for simulated input quantities[3] and hats ($\hat{\vec{s}}$) for estimated quantities recovered by reconstruction.

1.2.1 Simulation: Forward model

In the previous derivation of the map-making equation, the frequency dependence of the instrumental model H was neglected. For conventional imagers, this approximation is generally sufficient, since the instrumental response varies only weakly across the bandwidth[4]. However, as discussed in the description of the spectral imaging capabilities of bolometric interferometry (in Chapter 4), the frequency dependence plays a major role in shaping the QUBIC synthesized beam, as its multiple peaks move away from each other with λ, creating a much stronger, and exploitable, effect. The acquisition equation 1.2 must therefore be written as :

$$\tilde{\vec{d}} = H\tilde{\vec{s}} + \vec{n} = \int_{\nu_{min}}^{\nu_{max}} \mathcal{H}_\nu \tilde{\vec{s}}_\nu d\nu + \vec{n}. \tag{1.10}$$

For the numerical modelling, we have to discretise into frequency sub-bands to perform the integration, introducing the parameter N_{sub}, the number of sub-band :

$$\begin{aligned} \tilde{\vec{d}} &= \int_{\nu_{min}}^{\nu_{max}} \mathcal{H}_\nu \tilde{\vec{s}}_\nu d\nu + \vec{n} \\ &\approx \sum_{i=1}^{N_{sub}} \mathcal{H}_{\nu_i} \tilde{\vec{s}}_{\nu_i} \Delta_{\nu_i} + \vec{n}, \end{aligned} \tag{1.11}$$

with Δ_{ν_i} the bandwidth of the sub-bands associated with the frequency ν_i.

To ensure the validity of such an approximation, N_{sub} needs to be large enough. A deeper analysis of this parameter can be found in subsection 9.1.1.

[3] For real acquisition, $\tilde{\vec{d}}$ designates the observed TOD and $\tilde{\vec{s}}$ the true sky to reconstruct.

[4] Only the PSF's FWHM evolves linearly with λ, which effect is mild. The reconstruction being then performed at an effective FWHM.

The discretisation into N_{sub} sub-bands can be seen as a numerical trick to compute the integral over the frequency bandwidth. Computing the instrumental model at each frequency associated with the sub-bands is numerically expensive, and N_{sub} is therefore a purely numerical, unphysical parameter. For real data, we can consider $N_{sub} = \infty$, which is totally impractical for numerical modelling.

1.2.2 Reconstruction: Frequency Map-Making

This decomposition, introduced purely for simulation purposes, turns out to have a direct physical interpretation: the N_{sub} sub-band sky maps can be treated as independent unknowns and recovered jointly from the TOD. This is the key idea behind Frequency Map-Making. To make this reconstruction tractable, we introduce a second parameter $N_{\text{rec}} \leq N_{sub}$, the number of frequency maps to reconstruct, and reformulate the acquisition equation 1.11 as :

$$\begin{aligned} \vec{\tilde{d}} &= \sum_{i=1}^{N_{sub}} \mathcal{H}_{\nu_i} \vec{\tilde{s}_{\nu_i}} \Delta_{\nu_i} + \vec{n} \\ &= \sum_{i=1}^{N_{\text{rec}}} \left(\sum_{j=1}^{f_{sub}} \mathcal{H}_{\nu_{ij}} \Delta_{\nu_{ij}} \right) \vec{\hat{s}}_{\nu_i} + \vec{n}, \end{aligned} \tag{1.12}$$

where $f_{sub} \equiv N_{sub}/N_{\text{rec}}$, a hyperparameter defining the number of frequency sub-bands integrated into each reconstructed map. Here, $\vec{\hat{s}}_{\nu_i}$ denotes the estimated sky maps (unknowns of the reconstruction), with total size $(N_{\text{rec}}, N_{pix}, N_{stokes})$. The main advantage of such a method is that it does not affect the data acquisition (or the data simulation), which is always performed with equation 1.11. Including the extra N_{rec} dimension allows reconstructing multiple frequency maps inside the QUBIC bandwidth. Thus, the N_{rec} parameter can be adjusted during the data analysis process, depending on the needs. Still, N_{rec} suffers from limitations, as it cannot be arbitrarily large. We discussed in Chapter 4 that we should avoid overlapping peaks while discretising, imposing an upper limit of $N_{\text{rec}} \approx 5$ per focal plane. However, since the beam is completely frequency-dependent, even the smallest discretisation $N_{\text{rec}} = 2$ will create anti-correlations between the 2 reconstructed maps and a decrease in the signal-to-noise ratio in each sub-band, as well as an increase in the number of unknowns to reconstruct, resulting in a higher residual level. On the other hand, reconstructing multiple sky maps inside the physical bandwidth leads to higher spectral resolution, which will help to separate the CMB from its contaminants. In summary, increasing N_{rec} improves the component separation capabilities of QUBIC, but at the cost of noise in the maps due to anti-correlations introduced by spectral imaging.

In reality, one should also include frequency-space PSF in $\mathcal{H}_\nu$, but we don't. As a result, we reconstruct the frequency smoothed $\vec{\tilde{s}_{\nu_i}}$, just like with spatial resolution that we do not attempt to deconvolve.

1.2.3 Angular resolution

Fixed angular resolution reconstruction

Any telescope observing the sky reconstructs a blurred image of it, which depends on its angular resolution determined by the size of its beam. The blurring can be approximated by the convolution of the sky by a Gaussian with width equal to the FWHM of the beam. For QUBIC, the FWHM depends on the distance between horns, through : $\sigma_\nu = \frac{c}{\nu \Delta_h}$, with σ_ν the FWHM at frequency ν, c the speed of light and Δ_h the distance between two horns. As the frequency largely affects QUBIC's synthesized beam, spatial convolutions need to be treated carefully in the reconstruction algorithm. By noting C_ν the spatial convolution operator and $\vec{\tilde{s}}_\nu^{\,\infty}$ the sky at frequency ν and at "infinite" spatial resolution, the data acquisition model can be rewritten:

$$\vec{\tilde{d}} = \int_{\nu_{min}}^{\nu_{max}} \mathcal{H}_\nu C_\nu \vec{\tilde{s}}_\nu^{\,\infty} d\nu + \vec{n}. \tag{1.13}$$

Then, from the reconstruction point of view, each of the N_{rec} reconstructed maps corresponds to a combination of f_{sub} frequency sub-bands, each associated with a different angular resolution. This raises the question of the effective resolution at which the reconstructed maps should be defined. Since it is impossible to reconstruct information at an angular resolution higher than that contained in the data[5], the reconstructed map resolution is fundamentally limited by the instrumental resolution within the considered frequency range. For a given reconstructed sub-band, the highest achievable resolution therefore corresponds to the smallest synthesized beam, i.e. the highest frequency within the sub-band. Consequently, all frequency sub-bands contributing to a reconstructed map must be brought to a common angular resolution.

Therefore, we can define the resolution degradation operator $\mathcal{C}_{ij}$, which apply a Gaussian kernel with width :

$$\sigma_{ij} = \sqrt{\sigma_{\nu_i}^2 - \sigma_{\nu_j}^2}. \tag{1.14}$$

By introducing this operator in the reconstruction model, it ensures that the reconstruction of the sky is at the best resolution from QUBIC within each frequency sub-band. The reconstruction equation is then given by :

$$\vec{\tilde{d}} = \sum_{i=1}^{N_{\mathrm{rec}}} \left(\sum_{j=1}^{f_{sub}} \Delta_{\nu_{ij}} \mathcal{H}_{\nu_{ij}} \mathcal{C}_{ij} \right) \vec{\hat{s}}_{\nu_i} + \vec{n}. \tag{1.15}$$

Unfortunately, applying a convolution on the sky is an expensive operation, in particular if it is performed for each frequency sub-bands, and at each iteration of the numerical solver. Therefore, it considerably slows down the reconstruction, despite being necessary for the physical correctness and the realism of the method.

Free angular resolution reconstruction

To summarise, the effect of the beam width, modelled by the convolution operator, is included in the TOD simulation, and of course in the real data, but not in the

[5]Unless one has extremely high signal-to-noise ratio.

reconstruction process. The PCG solver therefore absorbs the individual sub-band beam responses into the pixel solution. As a result, the effective angular resolution of the reconstructed maps is not imposed a priori and may differ from one reconstructed band to another.

The reconstruction converges towards a map characterized by an effective angular resolution that can be approximated by the signal-weighted average of the sub-band resolutions:

$$\sigma_{\nu_i}^{\mathrm{rec}} = \frac{\sum_{j=1}^{f_{\mathrm{sub}}} \alpha_{\nu_j} w_{\nu_j} \sigma_{\nu_j}}{\sum_{j=1}^{f_{\mathrm{sub}}} \alpha_{\nu_j} w_{\nu_j}}, \tag{1.16}$$

where ν_j denotes the frequencies belonging to the reconstructed band ν_i, α_{ν_j} is the effective coverage defined as $H^{\,T} N^{-1} H$ and corresponding to the noise weight of each pixel, w_{ν_j} is a weighting factor, and σ_{ν_j} is the beam width associated with frequency ν_j. The weighting factor depends on the sky emission model, since the relative contribution of the different sky components varies across the frequency band. For instance, thermal dust emission increases with frequency, giving a larger weight to the highest-frequency sub-bands when averaging the angular resolution. The weights are therefore derived from the spectral energy distributions (SEDs) of the considered sky components.

To verify the validity of this hypothesis, several metrics have been used. The simplest and most satisfying one consists in running the algorithm without adding the instrumental and photon noise contribution in simulated data. Doing so, with an unbiased method, we are expected to reconstruct perfectly the input maps. Then, any small difference between the true and computed reconstructed resolutions will be visible as significant residuals. The result is shown in Figure 1.1.

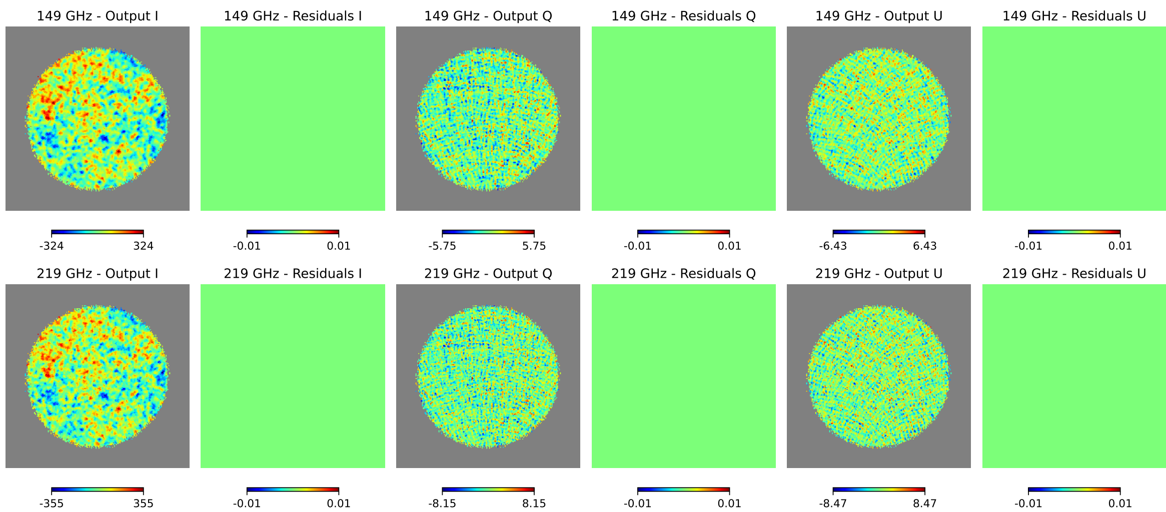

Figure 1.1: Reconstructed and residual maps in noiseless case. Residual are consistent with 0, showing that reconstruction is unbiased.

We can also use the noise profile in residual maps. Indeed, as the number of photons is split into two maps for polarisation, with respect to intensity one, we expect a noise ratio of $1/\sqrt{2}$ between the two. Observing this ratio between reconstructed maps is a good indication of an unbiased noise repartition. Figure 1.2 exhibits that noise's

RMS versus angular distance to the centre of the patch in residual maps is identical between I and $QU/\sqrt{2}$. The small fluctuations visible close to the centre are due to statistical fluctuation, as RMS is computed along fewer pixels closer to the centre than at the border. This result shows a consistent noise distribution between the 3 Stokes parameters.

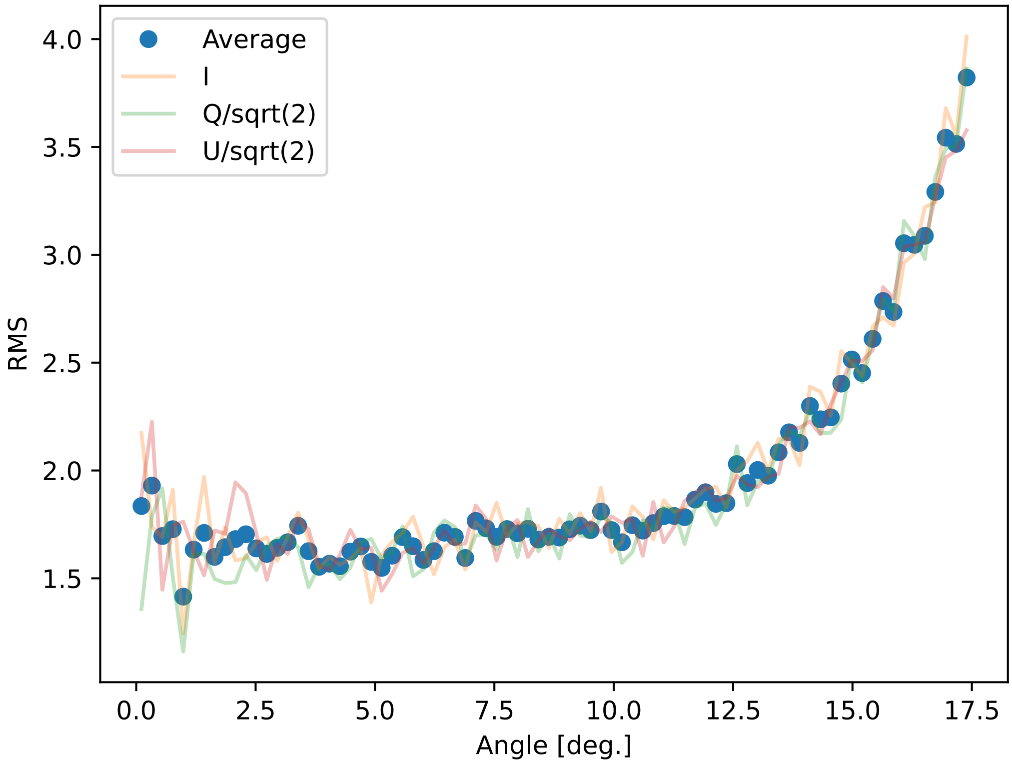

Figure 1.2: RMS of noise in the residual maps depending on their angular distance to the centre of the patch.

Finally, we can manually fit the resolution for which the difference between the reconstructed maps and the true maps at infinite resolution, convolved at this resolution, is the lowest. This is equivalent to minimising the function :

$$\chi^2(\sigma_{rec}) \propto \left(s_{rec} - C(\sigma_{rec}) s_{true}^{\infty}\right)^2 , \tag{1.17}$$

The result is presented in Figure 1.3, where the expected computed value for the reconstructed FWHM is very close to the one minimising the most the difference between input and reconstruction. The small difference being explained by the random noise in the reconstructed map, but is small enough to be confident in our computation.

This approach is currently under investigation and remains less robust than explicitly including the convolution operator in the reconstruction model. Nevertheless, within a controlled simulation framework, it provides a computationally efficient alternative that allows rapid exploration of the QUBIC map-making parameter space.

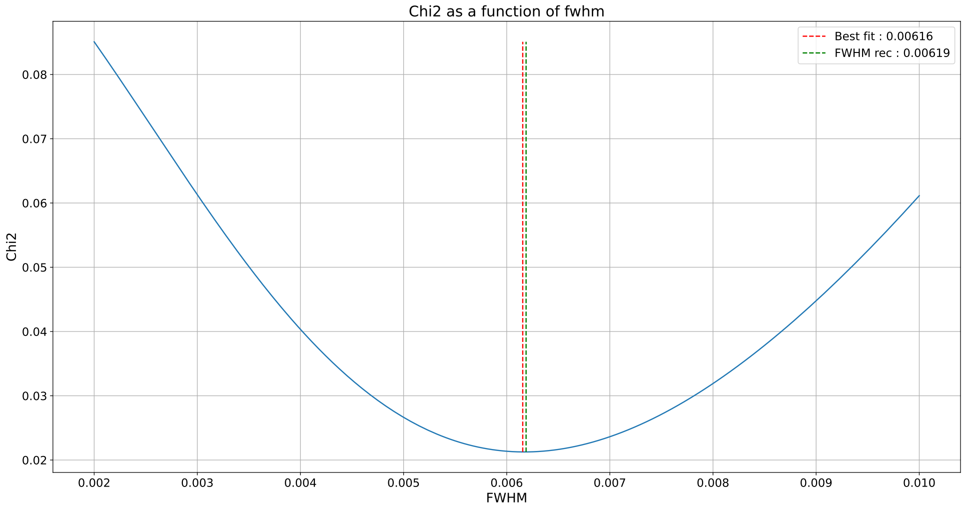

Figure 1.3: Fitting of the reconstructed FWHM for the 150 GHz map.

This simplification proved particularly useful for the analyses presented in this thesis. A detailed comparison between reconstructions performed with and without the convolution operator is presented in Chapter 9. Although the method performs satisfactorily in simulations, its applicability to real data remains uncertain. In particular, instrumental systematics, calibration uncertainties, and imperfect knowledge of the frequency response may alter the effective frequency mixing within reconstructed sub-bands, thereby affecting the predicted angular resolution. Further investigations are therefore required before this approach can be considered for the analysis of observational data.

1.2.4 Preconditioning

The main advantage of the PCG algorithm is the possibility of using a preconditioner to accelerate the convergence of the Conjugate Gradient method. The preconditioner is a symmetric positive-definite matrix M, which is applied to Eq. 1.8 as :

$$M^{-1}A\vec{x} = M^{-1}\vec{b}. \tag{1.18}$$

The matrix M should satisfy two requirements: it must not be expensive to invert and provide a good approximation of A. These two objectives are generally in competition. Indeed, the closer M is to A, the more costly it becomes to apply its inverse. The ideal preconditioner would be $M = A$, for which convergence is achieved in a single iteration.

In our implementation, the preconditioner is chosen as :

$$M = \mathrm{diag}(H^T H), \tag{1.19}$$

whose inverse can be applied at negligible computational cost. Although this approximation remains simplistic, since it neglects both the noise weighting and the cross-band correlations, it nevertheless provides a noticeable improvement in convergence speed, as illustrated in Figure 1.4.

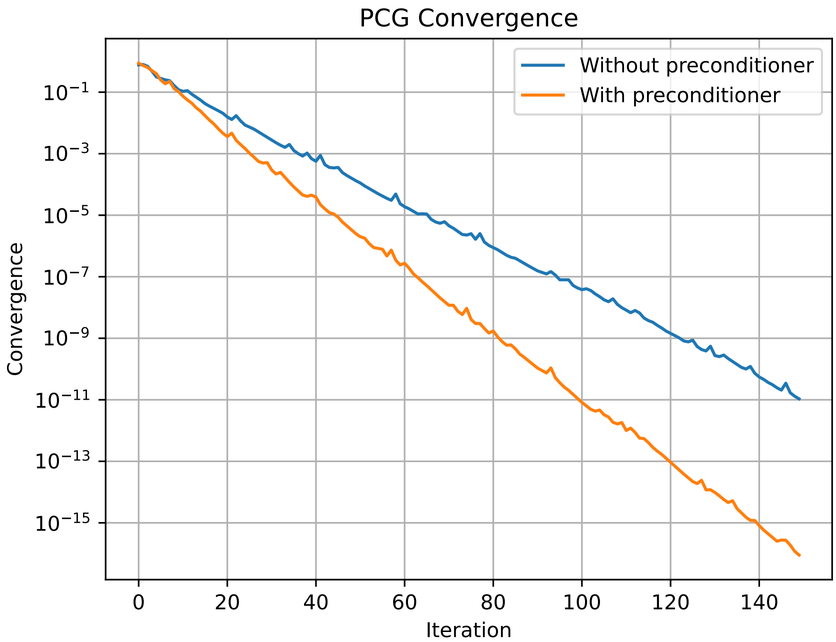

Figure 1.4: Comparison of the convergence against the PCG iteration with and without preconditioner.

The convergence at iteration k is quantified by the relative residual :

$$\epsilon_k \equiv \frac{|r_k|_2}{|b|_2} = \frac{|b - Ax_k|_2}{|b|_2}, \tag{1.20}$$

where r_k denotes the residual vector at iteration k. A stopping criterion is defined by requiring :

$$\epsilon_k < \text{tol}. \tag{1.21}$$

No dedicated study has yet been performed to determine an optimal value of tol. In practice, we adopt a sufficiently small tolerance together with a reasonably large maximum number of iterations, depending on the problem under consideration.

1.2.5 External data

The map reconstruction performed with the QUBIC synthesized beam suffers from an important edge effect. When observing regions close to the boundary of the patch, some secondary peaks of the synthesized beam fall outside the observed area, while others remain inside the well-sampled region. As a consequence, the signal measured near the edges depends partly on sky regions that are poorly constrained by QUBIC observations. This effect leads to higher degeneracy, and by extension, an increase in the reconstruction noise and a degradation of the map quality near the patch boundaries.

To mitigate this limitation, external observations are incorporated into the reconstruction. In practice, frequency maps from Planck are used to provide additional information outside the QUBIC patch, improving the constraints on emission surrounding the observed region.

The external observations, denoted by $\vec{d}_{\text{ext}}$, are incorporated into the data vector by extending the observation model as :

$$\vec{d}_{\text{tot}} = \begin{bmatrix} \vec{d}_{\text{QUBIC}} \\ \vec{d}_{\text{ext}} \end{bmatrix}. \tag{1.22}$$

The main modification concerns the acquisition operator, which must account simultaneously for QUBIC and external observations. The total acquisition operator can be written as

$$H_{\text{Tot}} = \begin{pmatrix} H^{\nu_1}_{\text{QUBIC}} & H^{\nu_2}_{\text{QUBIC}} & \cdots & H^{\nu_{N_{\text{rec}}}}_{\text{QUBIC}} \\ H^{\nu_1}_{\text{ext}} & 0 & 0 & 0 \\ 0 & H^{\nu_2}_{\text{ext}} & 0 & 0 \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & H^{\nu_{N_{\text{rec}}}}_{\text{ext}} \end{pmatrix} = \begin{pmatrix} H_{\text{QUBIC}} \\ H_{\text{ext}} \end{pmatrix} \tag{1.23}$$

The operators $H^{\text{QUBIC}}_{\nu_i}$ correspond to the QUBIC acquisition model for the reconstructed sub-band centred at frequency ν_i. In contrast, $H^{\text{ext}}_{\nu_i}$ is a much simpler operator that directly reads the corresponding external frequency map. Consequently, the external observations provide an additional constraint on each reconstructed sub-band while preserving the full QUBIC instrumental model. In practice, the external information is not taken directly from Planck frequency maps. Instead, we use the component maps obtained from the Planck component-separation analysis [12]. These maps are combined with the corresponding mixing matrix to reconstruct the expected emission at the QUBIC sub-band frequencies. This procedure ensures that the external information is expressed in the same frequency basis as the QUBIC reconstruction, allowing the two datasets to be combined consistently. Particular care is taken to avoid introducing external information within the QUBIC observed region, to avoid having possible Planck systematics[6]. The external constraints are therefore applied only outside the QUBIC patch, where they compensate for the lack of observational information and mitigate the edge effects induced by the synthesized beam. As a result, the reconstruction inside the observed region remains driven by QUBIC data, while the external maps provide the information required beyond the patch boundaries.

The inverse noise covariance matrix is also modified to account for noise associated with external data :

$$N^{-1} = \begin{bmatrix} N^{-1}_{\text{QUBIC}} & 0 \\ 0 & N^{-1}_{\text{ext}} \end{bmatrix}. \tag{1.24}$$

According to these modifications, the cost function to minimise can be rewritten as :

[6] Planck's information is added during the fit of cosmological and foregrounds parameters for the cross-spectra component separation introduced in Chapter 7. Therefore, we are not including Planck information inside the patch to not add two times the same information.

$$
\begin{aligned}
\chi^2_{\text{Tot}} &= \left(\vec{d} - \tilde{\vec{d}}\right)^T \cdot N^{-1} \cdot \left(\vec{d} - \tilde{\vec{d}}\right) \\
&= \left(\vec{d} - H_{\text{Tot}}\vec{s}\right)^T \cdot N^{-1} \cdot \left(\vec{d} - H_{\text{Tot}}\vec{s}\right) \\
&= \begin{bmatrix} \vec{d}_{\text{QUBIC}} - \tilde{\vec{d}}_{\text{QUBIC}} \\ \vec{d}_{\text{ext}} - \tilde{\vec{d}}_{\text{ext}} \end{bmatrix}^T \cdot \begin{bmatrix} N^{-1}_{\text{QUBIC}} & 0 \\ 0 & N^{-1}_{\text{ext}} \end{bmatrix} \cdot \begin{bmatrix} \vec{d}_{\text{QUBIC}} - \tilde{\vec{d}}_{\text{QUBIC}} \\ \vec{d}_{\text{ext}} - \tilde{\vec{d}}_{\text{ext}} \end{bmatrix} \\
&= \left(\vec{d}_{\text{QUBIC}} - H_{\text{QUBIC}}\vec{s}\right)^T \cdot N^{-1}_{\text{QUBIC}} \cdot \left(\vec{d}_{\text{QUBIC}} - H_{\text{QUBIC}}\vec{s}\right) \\
&\quad + \left(\vec{d}_{\text{ext}} - H_{\text{ext}}\vec{s}\right)^T \cdot N^{-1}_{\text{ext}} \cdot \left(\vec{d}_{\text{ext}} - H_{\text{ext}}\vec{s}\right) \\
&= \chi^2_{\text{QUBIC}} + \chi^2_{\text{ext}}
\end{aligned}
$$

1.3 Results

Now that we detailed the Frequency Map-Making algorithm, we can start looking at its results. In this section, we present reconstructed maps using FMM for few cases (a full analysis is developed in Chapter 9), as well as characterisation of the algorithm. This includes an estimation of the numerical resources and the need to parallelise to run the algorithm, an analysis of the structure of noise in the maps, and the effect of splitting the physical bands of QUBIC in the maps.

1.3.1 Simulation setup

In order to run the FMM algorithm from simulating data to reconstruct sky maps, we need to define some concepts.

Scanning Strategy

The scanning strategy consists in defining where the telescope points to and when. It is a tool used to mitigate instrumental systematics and atmospheric fluctuations, by scanning the same part of the sky multiple times. For the simulation in this thesis, it was chosen not to use a realistic scanning strategy, but a random scanning strategy. It means that at each time, the instrument will point at a random position inside the patch. The goal of this method is to achieve a good coverage of the sky patch with a reasonable number of points. It allows for a simpler instrumental model and a shorter dataset, both of which benefit the numerical computation, and should not affect the result of the reconstruction (but, with a realistic scanning strategy, it would only take much more pointing to achieve the same coverage level of the patch). Additionally, the location on the sky matters for a scientific survey. For CMB experiments, the main criterium is a region with low astrophysical emissions, meaning that we can't observe close to the Milky Way centre. For all figures inside this thesis, the simulated instrument observes the expected scientific patch of QUBIC experiment, described in subsection 2.2.3.

1.3.2 Maps

By applying the map-making on a simulated sky[7] composed of CMB and Dust, we can reconstruct sky maps, as shown in Figure 1.5 and 1.6. These plots and a majority of plots within this thesis, follow the same format as the two figures presented here. The left column represents the input map, or the true map, used to simulate the data. The central column is the output map, the one reconstructed by PCG. Finally, the right column is the residual map, the difference between true and reconstructed maps. The residual map is supposed to include only noise for a perfect reconstruction. Figure 1.5 shows the reconstruction into $N_{\text{rec}} = 4$ frequency bands of the Intensity sky maps using the DB configuration. Each line corresponds to one frequency sub-band reconstructed, in μK_{CMB} units. On the other hand, Figure 1.6 shows reconstruction for the two polarisation parameters Q and U, for a 150 GHz band, in the case $N_{\text{rec}} = 2$. The first line corresponding to the Q map and the second the U map.

1.3.3 Numerical resources: memory and computational time

One of the main limitations of the Frequency Map-Making algorithm is its memory usage, as well as the computation time.

Memory usage

The core complexity lies in H operator, which can be seen as a sparse matrix simulating the instrument (a detailed description of this operator is given in Chapter 8). The number of peak is given by the maximum diffraction order considered $k_{\max}$, see 4.1.2 for more details. This matrix, for one frequency band, has a shape of $(N_{det} \times N_{\text{pointings}}, N_{peak})$. Considering that the matrix is composed of float64, written 64 bits $=$ 8 bytes. Thus, for $N_{det} = 992$, $N_{\text{pointings}} = 10000$, and $N_{peak} = (2k_{\max}+1)^2 = 49$ with $k_{\max} = 3$, the resulting memory used is then $10000 \times 992 \times 49 \times 8 = 3.889 \times 10^9$ bytes $= 3.622$ GiB[8]. So, for $N_{sub} = 40$, we need ~ 144.9 GiB of RAM only to store the operator H. It is important to note that the value given as an example are far from being extreme. It explains the need for a random scanning strategy, as explained in the part 1.3.1. Additionally, a complete study of the minimal parameter value is discussed in Chapter 9, to save as much memory as possible. In reality, the reconstruction can't be done with large N_{sub}, $N_{\text{pointings}}$ or $k_{\max}$ values, as a large H results in a very slow reconstruction with PCG. Therefore, it is important for realism to simulate data with the largest value possible (in particular N_{sub} and $k_{\max}$, as they are equal to ∞ in real acquisition). In order to achieve that, I designed a sequential data simulation, where multiple TOD with small $N_{\text{pointings}}$ are created and concatenate into a large one. This method allows for a TOD simulation with large parameters while being still runnable with a reasonable amount of memory.

[7] simulated using PySM3 framework : `https://github.com/galsci/pysm`.

[8] 1 GiB = 1024 MiB = 1024^2 KiB = 1024^3 bytes.

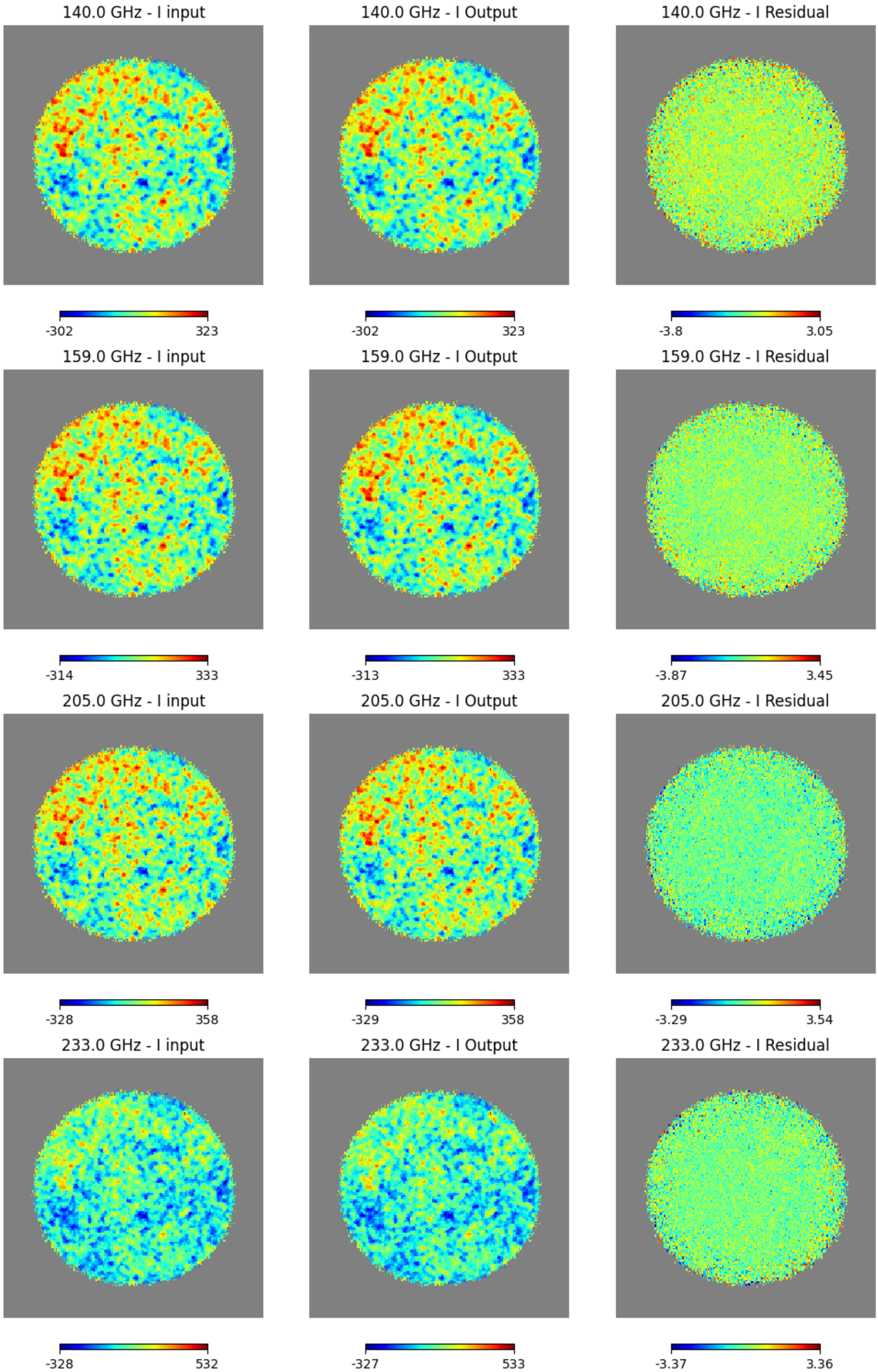

Figure 1.5: Intensity maps of the sky with $N_{\mathrm{rec}} = 4$. For this figure, the noise level has been adjusted for illustration purpose, and does not represent the real noise expected for QUBIC instrument.

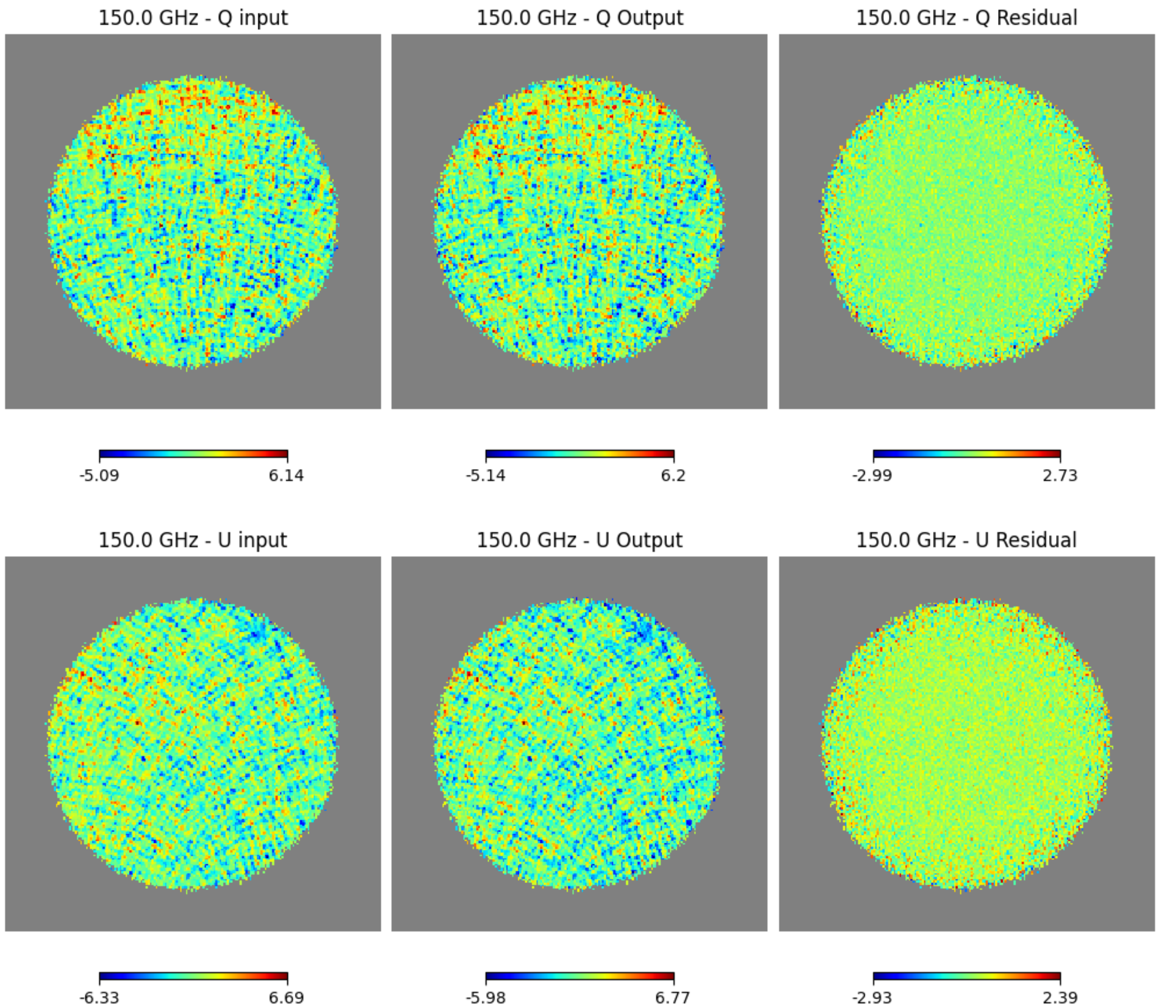

Figure 1.6: Q and U maps of the sky, with $N_{\mathrm{rec}} = 2$. For this figure, the noise level has been adjusted for illustration purpose, and does not represent the real noise expected for QUBIC instrument.

Computational time

The computational time to run the FMM algorithm is its second limitation alongside memory. As expected considering the size of H, it is very expensive to apply this operator on a sky to simulate data. Unfortunately, the iterative reconstruction through PCG implies applying H at each iteration, leading to a significant computational cost. Indeed, considering $N_{det} = 992$, $N_{\text{pointings}} = 1000$, $N_{peak} = (2k_{\max} + 1)^2 = 9$ with $k_{\max} = 1$ and $N_{sub} = 20$, each PCG iteration takes ~ 15 sec for an operator H with a memory footprint of ~ 6.65 GiB. These parameters are relatively small compared to those used in the previous section, in order to run on a simple laptop. To achieve this reasonable time per iteration, given that PCG requires ~ 100 iterations to converge, parallelisation and optimisation are included. In particular, the most expensive part of H, which computes the coordinate change between sky positions and the Cartesian instrumental frame, is written in Fortran and parallelised with OpenMP[9]. OpenMP (Open Multi-Processing) is an API for thread-based parallelism on a shared-memory machine, allowing multiple CPU cores to work on the same data simultaneously. Additionally, MPI (Message Passing Interface), through mpi4py[10], is used for the Python part of the simulation software, distributing the time samples and the detectors across processes. While OpenMP is active even on a laptop, MPI is exploited when the code is run on an HPC cluster, where multiple nodes are requested.

1.3.4 Noise structure

It is important to understand the noise profile in the reconstructed maps. As visible in Figures 1.5 and 1.6, the noise in the residual maps is not uniform across the sky patch. The feature is due to the coverage of the patch being inhomogeneous. Indeed, due to the scanning strategy we are using to simulate the data acquisition, it is expected to observe the inner part of the patch more often[11]. The coverage, corresponding to the number of time all pixels are observed by the instrument, is plotted in Figure 1.7. It is computed by applying H^T, the transpose of the instrumental model H, on a uniform sky.

From a statistical point of view, it is expected to have the worst reconstruction in the regions where the coverage is smaller, as we spend less time to observe the outer parts. A way to show that is by plotting the noise profile regarding the angular distance to the centre of the patch. The result is presented in Figure 1.8.

This figure exhibits multiple interesting features. First, looking at the RMS of the residual maps, we can see a plateau where the RMS is relatively constant, then, it starts rising (approximately where we are at a distance equivalent to the distance between peaks from the edge of the patch). Looking at the coverage, it seems less flat in the inner part of the patch, but exhibits the same rising when reaching the outer regions. The ratio of $1/\sqrt{2}$ between Intensity map and Polarisation maps is purely statistical, the number of photons being divided by 2 due to the 2 Polarisation maps in comparison

[9] `https://www.openmp.org/`

[10] `https://pypi.org/project/mpi4py/`

[11] It is important to note that with a more realistic scanning strategy, this effect will be more important than with the random one. In addition, the coverage will also be less uniform for the same number of time sample.

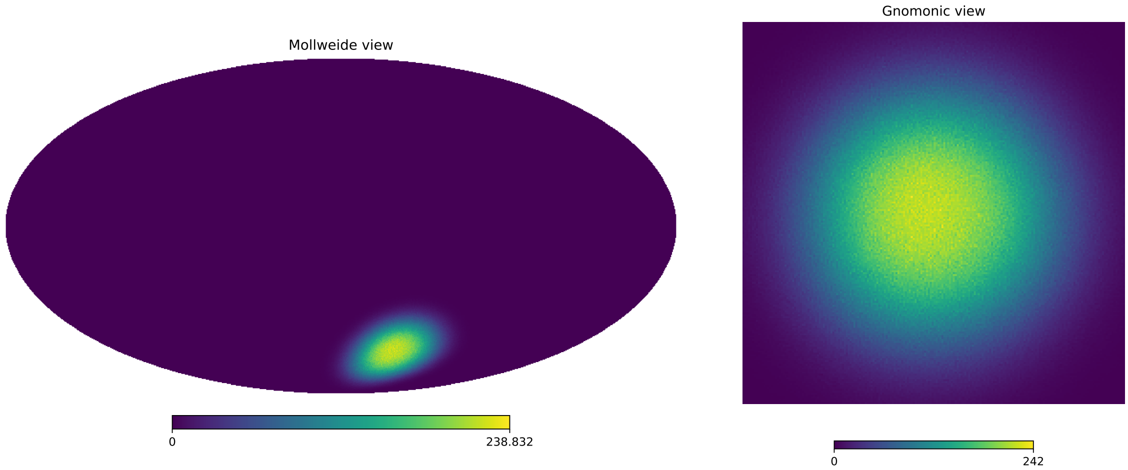

Figure 1.7: Coverage of QUBIC sky patch in Mollview projection (left) and Gnomview projection (right).

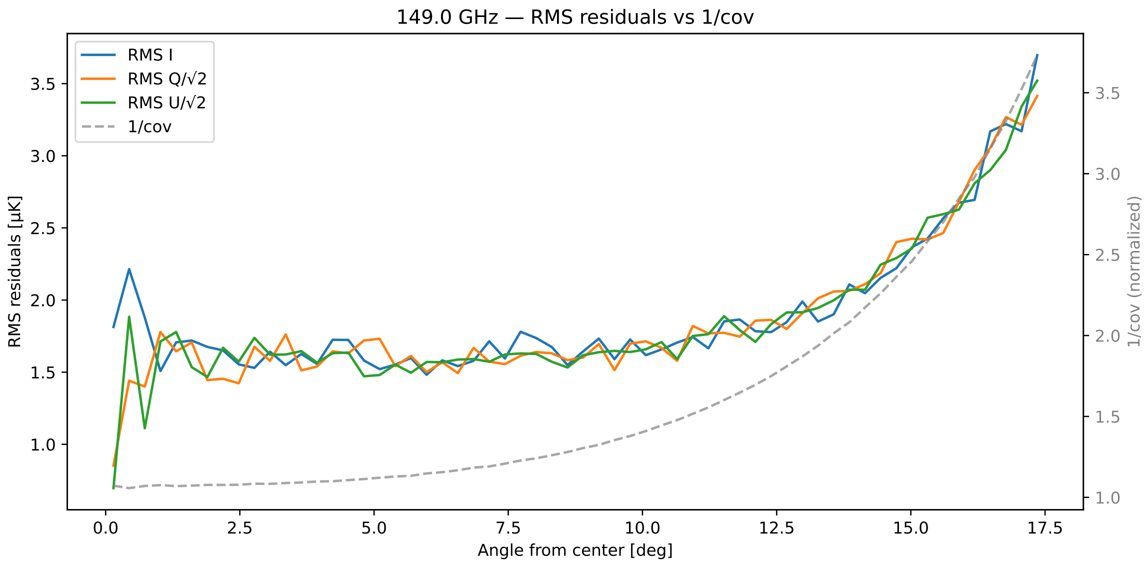

Figure 1.8: RMS of the residual map with the angular distance to the centre of the patch. Blue points show the average between I, Q and U. Grey dash line shows the inverse of the normalised coverage.

with Intensity. Observing $RMS_I \sim RMS_{QU}/\sqrt{2}$ means a correct noise repartition in the residual maps. For an imager, one typically expects a direct correlation between 1/coverage and the RMS of the noise in the residual maps. Here, the inverse of the coverage follows the noise profile only in the outer part of the patch, but not in the central regions. For QUBIC, the behaviour appears different, as the RMS of the noise follows 1/coverage only in the outer regions of the patch. This is still ongoing work and requires further investigation of the noise structure in the maps, to determine whether this effect originates from the simulation software or is an intrinsic feature of bolometric interferometry and spectral imaging.

1.3.5 Sub-bands Correlations

For a classical imager, the sky is observed at different frequencies, using a different detector chain for each of them. It results that the noise in the reconstructed maps is independent between frequency channels : $Cov(\nu_i, \nu_j) \approxeq 0$ for $i \neq j$[12]. It is different for a Bolometric Interferometer, as it is possible to reconstruct multiple maps from data taken by the same detectors. Indeed, we are extracting the spectral information from the shape of the interference pattern on the focal plane. Since several reconstructed sub-bands contribute simultaneously to the same detector samples, a positive fluctuation assigned to one sub-band must be compensated by a negative fluctuation in neighbouring sub-bands in order to reproduce the same data. It means that reconstructed maps are anti-correlated with each other : $Cov(\nu_i, \nu_j) < 0$ for $i \neq j$. To understand this behaviour in more depth, it is important to note that close sub-bands are more anti-correlated than distant ones because the synthesized beam changes only slightly with frequency. Consequently, neighbouring sub-bands produce very similar signatures in TOD and are harder to disentangle during the reconstruction. Figure 1.9 is illustrating these two specificities for BI. We are reconstructing 3 maps within one physical band, and we compare the correlation, average over pixels, between reconstructed sub-maps for each Stokes parameters. The anti-correlation between the sub-bands ν_0, ν_1 and ν_2 is visible. Also, as explained just above, anti-correlation is higher for neighbouring sub-bands. We don't expect any correlation between Stokes parameters for a perfect polariser and Half-Wave Plate, as confirmed in the figure.

Finally, the higher N_{rec} is (so, the more maps we are trying to reconstruct), the closer the shape of the beam between adjacent sub-bands will be, leading to stronger anti-correlations and a degradation of the noise performance in the reconstructed maps. This is an intrinsic limitation to spectral imaging, when we can't distinguish the signal between narrow sub-bands.

Conclusion

Through this chapter, we explained how spectral imaging can be used to reconstruct multiple frequency band within one physical band, defining the core of Frequency Map-Making. The main goal behind this algorithm is to improve the spectral resolution and the amount of frequency information, which is the key point to improve component

[12]This assumption could be broken by considering atmospheric noise.

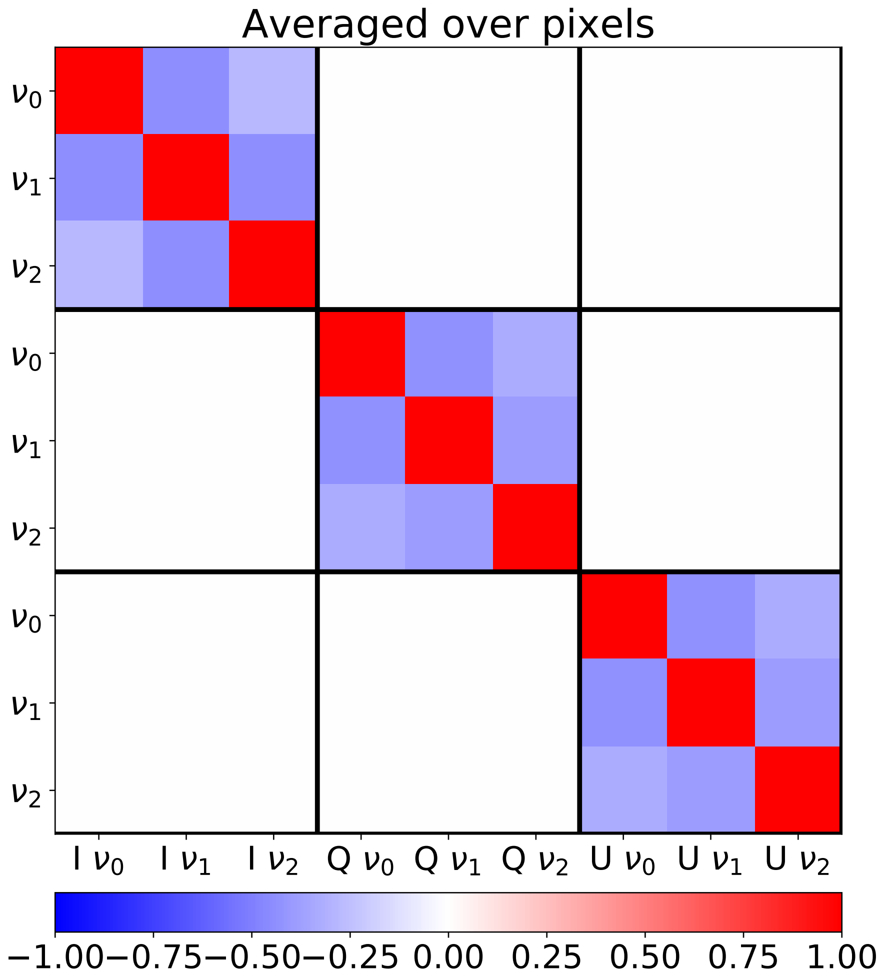

Figure 1.9: Correlation matrix over the pixels for 3 reconstructed maps within the physical band, for the 3 Stokes parameters. Taken from [6].

separation. The component separation method used on the reconstructed frequency maps is presented in Chapter 7, while the hyperparameters effects and TOD realism criteria are discussed in Chapter 9.